The University of the South Pacific

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School of Information Technology, Engineering, Mathematics and Physics

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CS211: Computer Organisation

Test 2 – Semester 1, 2023

Time Allowed: 50 mins

Total Marks: 30 (15% of final grade)

Name: _____ Student ID: _____

INSTRUCTIONS:

- There are 6 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

1. Data path cycle is the heart of CPU. Explain data path cycle. (2 marks)

   Ans: The process of running two operands through the ALU and storing the result is called the data path cycle.

2. You are working on a microarchitecture design for a new processor, and your team is considering implementing branch prediction to improve performance. You have two options for the branch predictor:
   Option 1: A static branch predictor that always predicts the branch will not be taken.
   Option 2: A dynamic branch predictor that uses a two-bit counter to keep track of the branch's behavior history.
   Which option do you recommend and why? (3 marks)
   Answer: I would recommend implementing the dynamic branch predictor (Option 2). While a static branch predictor is simple to implement and can sometimes be effective, it does not take into account the behavior history of the branch. A dynamic branch predictor, on the other hand, uses the branch history to make a more accurate prediction, improving performance. By keeping track of the history of the branch behavior, the dynamic branch predictor can adjust its prediction to match the branch's behavior, allowing for faster and more efficient execution of code. Therefore, Option 2 would be a better choice for improving performance in the microarchitecture design.

3. Suppose you are working with a team to design an instruction set architecture (ISA) for a new processor. Your team is considering adding a new instruction to the ISA to improve performance for a specific application. The instruction will be used to perform a common operation that occurs frequently in the application. Your team has come up with two possible designs for the instruction:
   Design 1: The new instruction will take two operands from the general-purpose register file, add them together, and store the result back in one of the register files. The instruction will take 4 clock cycles to complete.
   Design 2: The new instruction will take one operand from the general-purpose register file and one operand from the memory, add them together, and store the result back in the general-purpose register file. The instruction will take 6 clock cycles to complete.
   Which design do you recommend your team implement in the ISA and why? (4 marks)
   Design 1 is recommended because it is faster than Design 2. Taking operands from the register file is faster than taking operands from memory because register files are located on-chip, while memory is located off-chip. Accessing memory takes more time than accessing the register file. Design 1 takes 4 clock cycles to complete, while Design 2 takes 6 clock cycles to complete. Therefore, Design 1 is more efficient and should be implemented in the ISA.

4. Assume the following instructions are available for 0-address (stack-based) and 1-address machines: ADD, SUB, MUL, DIV, LOAD, STORE, PUSH, POP. Memory addresses like A, B, C, etc. are used as operands. Each instruction is made up of 24 bits operand and 8 bits of opcode.

    a. Write instructions for a 0-address (stack) machine to compute X = (A-B)/C+D/E. (3 marks)

    ```
    PUSH  A
    PUSH  B
    SUB
    PUSH C
    DIV
    PUSH  D
    PUSH  E
    DIV
    ADD
    POP   X
    ```

    b. How many bits does 0-address machine need to compute X? (1 mark)

    10 X 8 + 24 X 6 = 224

    c. Write instructions for a 1-address machine to compute X = (A-B)/C+D/E. (3 marks)

    ```
    LOAD  D
    DIV   E
    STORE T
    LOAD  A
    SUB   B
    DIV C

    ADD   T
    STORE X
    ```

    d. How many bits does 1-address machine need to compute X? (1 mark)

    8 X 32 = 256 bits

5. Consider the case of a processor with an instruction length of 12 bits and with 32 general-purpose register so the size of the address fields is 5 bits. Is it possible to have instruction encodings for the following? (3 marks)

- 3 two-address instructions
- 30 one-address instructions
- 45 zero-address instructions

ANS:

00 A B

10 A B

11 00000 B
.
.
11 11101 B

11 11110 00000
.
.
11 11110 11111
11 11111 01100

Design is possible.

1 mark for each criterion

6. A function F is defined in the truth below below:

x	y	z	F(x,y,z)
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

a. Write the disjunctive normal form (DNF) for the above function F. (2 marks)
b. Simply the above DNF expression in its simplest form. (3 marks)
c. Draw the circuit diagram for the simplest form in part b) above. (2 marks)
d. Use the above truth table to draw the three variable multiplexer chip for the function F. (2 marks)

ANS:

a) $F(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz$

b)

$= \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz$

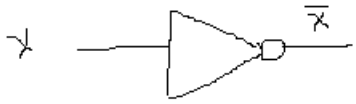
$\bar{x}\overline{y(\bar{z} + z)} + \bar{x}y(\bar{z} + z)$

$\bar{x}\bar{y} + \bar{x}y$

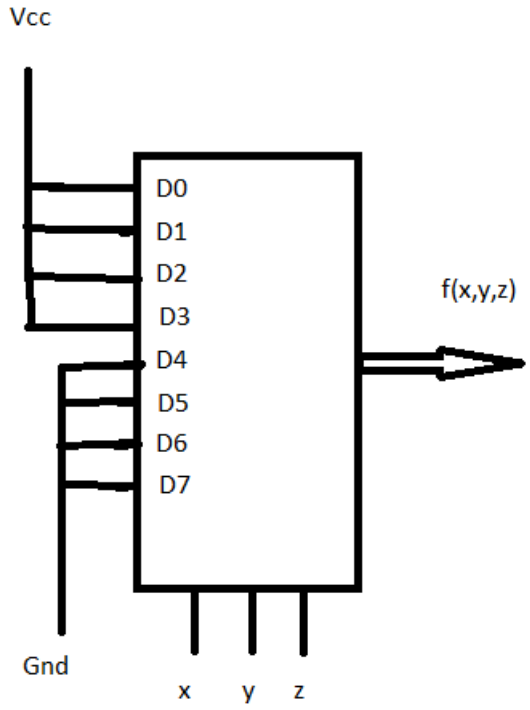
$\overline{x(\bar{y} + y)}$

$\bar{x}$

c)



d)



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